

3. Docker Storage

Demonstrate:

1. Loss of container-layer data after container deletion.
2. Permanent storage using a Docker named volume and bind mount

```
(kali㉿kali)-[~]
└─$ sudo docker rm -f container1 container2
[sudo] password for sanchi:
container1
container2

(kali㉿kali)-[~]
└─$ sudo docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
3b7138528a62   ubuntu   "bash"    5 hours ago   Exited (129) 44 seconds ago           test-container

(kali㉿kali)-[~]
└─$ sudo docker network rm mynetwork
mynetwork

(kali㉿kali)-[~]
└─$ sudo docker network create mynetwork
387c554006549dd6934e176485396eab9c05fa9ba6c3fbf5b421204dfc585b2d

└─$ sudo docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
7ae91da805ef   bridge    bridge     local
52c151f927df   host      host       local
387c55400654   mynetwork bridge     local
116413220f62   none      null       local

(kali㉿kali)-[~]
└─$ sudo docker run -dit --name container1 --network mynetwork ubuntu bash
1dc29ab3e1948f57c6a147ba4e978376119677845cfdd437d1af62a62652ed85

(kali㉿kali)-[~]
└─$ sudo docker run -dit --name container2 --network mynetwork ubuntu bash
20561cb4e3a56588bc03add7ac99b2c8f1a847e5b7f36016508acacdb6fa5849

(kali㉿kali)-[~]
└─$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
20561cb4e3a5   ubuntu   "bash"    8 seconds ago   Up 7 seconds           container2
1dc29ab3e194   ubuntu   "bash"    16 seconds ago   Up 15 seconds          container1

└─$ sudo docker exec -it container1 bash
root@1dc29ab3e194:/# apt update
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [212 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Packages [11.2 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [578 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [489 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute/restricted amd64 Packages [189 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute/main amd64 Packages [1874 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 Packages [352 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Packages [20.1 MB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [505 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [743 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [14.6 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [338 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Packages [3306 B]
Fetched 26.0 MB in 12s (2172 kB/s)
24 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
root@1dc29ab3e194:/# apt install -y iputils-ping
Installing:
  iputils-ping

Installing dependencies:
  libcap2 libidn2-0 libunistring5 linux-sysctl-defaults

Summary:
  Upgrading: 0, Installing: 5, Removing: 0, Not Upgrading: 24
  Download size: 761 kB
  Space needed: 2677 kB / 56.0 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libcap2 amd64 1:2.75-10ubuntu2 [29.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libunistring5 amd64 1.3-2build1 [610 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 libidn2-0 amd64 2.3.8-4build1 [67.6 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 iputils-ping amd64 3:20250605-1ubuntu1 [47.1 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute/main amd64 linux-sysctl-defaults all 4.15ubuntu5 [6208 B]
Fetched 761 kB in 3s (225 kB/s)
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/DebianConfig/Dialog.pm line 79, <STDIN> line 5.)
```

```
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC entries checked: /etc/perl /usr/local/lib/x86_64-linux-gnu/perl/5.40.1 /usr/local/share/perl/5.40.1 /usr/lib/x86_64-linux-gnu/perl/5.40.1 /usr/share/perl5 /usr/lib/x86_64-linux-gnu/perl-base /usr/lib/x86_64-linux-gnu/perl/5.40 /usr/share/perl/5.40 /usr/local/lib/site_perl) at /usr/share/perl5/DebianConfig/Dialog.pm line 8, <STDIN> line 5.)
debconf: falling back to frontend: Teletype
Selecting previously unselected package libcap2:amd64.
(Reading database ... 7724 files and directories currently installed.)
Preparing to unpack .../libcap2_1%3a2.75-10ubuntu2_amd64.deb ...
Unpacking libcap2:amd64 (1:2.75-10ubuntu2) ...
Selecting previously unselected package libunistring5:amd64.
Preparing to unpack .../libunistring5_1.3-2build1_amd64.deb ...
Unpacking libunistring5:amd64 (1.3-2build1) ...
Selecting previously unselected package libidn2-0:amd64.
Preparing to unpack .../libidn2-0_2.3.8-4build1_amd64.deb ...
Unpacking libidn2-0:amd64 (2.3.8-4build1) ...
Selecting previously unselected package iputils-ping.
Preparing to unpack .../iputils-ping_3%3a20250605-1ubuntu1_amd64.deb ...
Unpacking iputils-ping (3:20250605-1ubuntu1) ...
Selecting previously unselected package linux-sysctl-defaults.
```

```
Preparing to unpack .../linux-sysctl-defaults_4.15ubuntu5_all.deb ...
Unpacking linux-sysctl-defaults (4.15ubuntu5) ...
Setting up libcap2:amd64 (1:2.75-10ubuntu2) ...
Setting up linux-sysctl-defaults (4.15ubuntu5) ...
Setting up libunistring5:amd64 (1.3-2build1) ...
Setting up libidn2-0:amd64 (2.3.8-4build1) ...
Setting up iputils-ping (3:20250605-1ubuntu1) ...
/var/lib/dpkg/info/iputils-ping.postinst: 8: setcap: not found
Processing triggers for procps (2:4.0.4-9ubuntu1) ...
procps: Applying updated sysctl configuration
sysctl: setting key "kernel.sysrq", ignoring: Read-only file system
sysctl: setting key "kernel.core_uses_pid", ignoring: Read-only file system
sysctl: setting key "net.ipv4.conf.default.rp_filter", ignoring: Read-only file system
sysctl: setting key "net.ipv4.conf.eth0.rp_filter", ignoring: Read-only file system
sysctl: setting key "net.ipv4.conf.lo.rp_filter", ignoring: Read-only file system
sysctl: setting key "net.ipv4.conf.default.accept_source_route", ignoring: Read-only file system
sysctl: setting key "net.ipv4.conf.eth0.accept_source_route", ignoring: Read-only file system
sysctl: setting key "net.ipv4.conf.lo.accept_source_route", ignoring: Read-only file system
sysctl: setting key "net.ipv4.conf.default.promote_secondaries", ignoring: Read-only file system
sysctl: setting key "net.ipv4.conf.eth0.promote_secondaries", ignoring: Read-only file system
sysctl: setting key "net.ipv4.conf.lo.promote_secondaries", ignoring: Read-only file system
sysctl: setting key "net.ipv4.ping_group_range", ignoring: Read-only file system
sysctl: setting key "fs.protected_hardlinks", ignoring: Read-only file system
sysctl: setting key "fs.protected_symlinks", ignoring: Read-only file system
```

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sysctl: setting key "vm.mmap_min_addr", ignoring: Read-only file system
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
root@1dc29ab3e194:/# ping container2
PING container2 (172.18.0.3) 56(84) bytes of data.
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=1 ttl=64 time=9.48 ms
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=2 ttl=64 time=0.102 ms
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=3 ttl=64 time=0.100 ms
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=4 ttl=64 time=0.136 ms
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=5 ttl=64 time=0.085 ms
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=6 ttl=64 time=0.115 ms
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=7 ttl=64 time=0.092 ms
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=8 ttl=64 time=7.75 ms
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=9 ttl=64 time=0.165 ms
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=10 ttl=64 time=0.105 ms
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=11 ttl=64 time=0.168 ms
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=12 ttl=64 time=0.111 ms
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=13 ttl=64 time=0.151 ms
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=14 ttl=64 time=0.092 ms
^C
  -- container2 ping statistics --
14 packets transmitted, 14 received, 0% packet loss, time 13240ms
rtt min/avg/max/mdev = 0.085/1.331/9.475/2.989 ms
root@1dc29ab3e194:/# exit
exit

```

```

$ sudo docker inspect -f '{{range.NetworkSettings.Networks}}{{.IPAddress}}{{end}}' container2
172.18.0.3

(kali@kali)~$ sudo docker exec -it container1 bash
root@1dc29ab3e194:/# ping 172.18.0.3
PING 172.18.0.3 (172.18.0.3) 56(84) bytes of data.
64 bytes from 172.18.0.3: icmp_seq=1 ttl=64 time=0.413 ms
64 bytes from 172.18.0.3: icmp_seq=2 ttl=64 time=0.115 ms
64 bytes from 172.18.0.3: icmp_seq=3 ttl=64 time=0.107 ms
64 bytes from 172.18.0.3: icmp_seq=4 ttl=64 time=0.096 ms
64 bytes from 172.18.0.3: icmp_seq=5 ttl=64 time=0.145 ms
^C
  -- 172.18.0.3 ping statistics --
5 packets transmitted, 5 received, 0% packet loss, time 4101ms
rtt min/avg/max/mdev = 0.096/0.175/0.413/0.120 ms
root@1dc29ab3e194:/# exit
exit

```

```

$ sudo docker network inspect mynetwork
[
  {
    "Name": "mynetwork",
    "Id": "387c554006549dd6934e176485396eab9c05fa9ba6c3fbf5b421204dfc585b2d",
    "Created": "2026-09-08T09:53:28.501399039-04:00",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.18.0.0/16",
          "Gateway": "172.18.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
  }
]

```

```

File Actions Edit View Help
{
  "ConfigFrom": {
    "Network": ""
  },
  "ConfigOnly": false,
  "Containers": {
    "1dc29ab3e1948f57c6a147ba4e978376119677845cfd437d1af62a62652ed85": {
      "Name": "container1",
      "EndpointID": "35be79c5766287a7837da63de65ed1c8296f008e02ce87529afa8d352b69391e",
      "MacAddress": "ae:a7:4f:4e:c7:c1",
      "IPv4Address": "172.18.0.2/16",
      "IPv6Address": ""
    },
    "20561cb4e3a56588bc03add7ac99b2c8f1a847e5b7f36016508acacdb6fa5849": {
      "Name": "container2",
      "EndpointID": "6ec18e9ee9b98512073e4c15e200b9da783a04ffb3e3ac10717ba981605046db",
      "MacAddress": "ce:eb:89:8a:64:59",
      "IPv4Address": "172.18.0.3/16",
      "IPv6Address": ""
    }
  },
  "Options": {},
  "Labels": {}
}
]

```

```

$ sudo docker volume ls
DRIVER    VOLUME NAME
local     myvolume

(kali@kali)-[~]
$ sudo docker run -it --name new-volume-container \
-v myvolume:/data ubuntu bash
root@b07285dfedf5:/# cat /data/volume-data.txt
This data is stored in Docker named volume
root@b07285dfedf5:/# exit
exit

(kali@kali)-[~]
$ mkdir -p ~/docker-bind-data

```

```

$ sudo docker run -it --name bind-container \
-v ~/docker-bind-data:/data ubuntu bash
root@86293ae93565:/# echo "This data is stored using a bind mount" > /data/bind-data.txt
root@86293ae93565:/# cat /data/bind-data.txt
This data is stored using a bind mount
root@86293ae93565:/# exit
exit

(kali@kali)-[~]
$ sudo docker rm bind-container
bind-container

(kali@kali)-[~]
$ cat ~/docker-bind-data/bind-data.txt
This data is stored using a bind mount

```